

AI META PROMPTING AS COGNITIVE SCAFFOLDING IN TEACHING WRITING

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Abstract. *Rapid emergence of generative artificial intelligence (AI), is transforming the landscape of writing instruction. While current applications often emphasize surface-level uses like grammar correction or paraphrasing, this study proposes a deeper pedagogical integration: conceptualizing AI as metacognitive and dialogic scaffold capable of supporting higher-order thinking. Drawing on Cognitive Load Theory, Sociocultural Theory, Frameworks of Dialogic Learning, and Critical Pedagogy—the research situates AI as a non-human dialogic agent that can interactively engage students within their Zone of Proximal Development. Using a Design-Based Research methodology, the study implemented an AI-integrated instructional model in the course of English as a Second Language (ESL). The intervention combined Socratic instruction in strategic prompting with guided use of AI across writing stages. Collected data and findings indicate that AI, when framed as Socratic dialogue partner, can reduce extraneous cognitive load and increase engagement with germane cognitive processes, particularly in ideation and revision. Students reported greater clarity in argument structure and enhanced rhetorical awareness. Importantly, personalized AI feedback was perceived as a motivating guide contingent on students' ability to co-direct the interaction through intentional prompting strategies. The study contributes a theoretical and pedagogical framework for AI-responsive writing instruction and raises critical questions about ethical literacy and the evolving role of educators in AI-augmented learning environments.*

Key words: *AI-supported writing practices, meta prompt, metacognitive reflection, metacognitive scaffold, design-based research methodology, Cognitive Load Theory*

1. INTRODUCTION

The rapid emergence of generative artificial intelligence (AI) technologies, such as large language models (LLMs), is reshaping the epistemic and pedagogical foundations of writing instruction in higher education. The introduction of ChatGPT, Claude, and Gemini, to name a few, is at present making a profound shift in higher education, including how writing is conceived, taught, and produced. Despite the proliferation of research on AI capabilities, much of the scholarly attention has focused on AI as a tool

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for mechanistic enhancement, improving sentence-level accuracy, rephrasing awkward constructions, or summarizing the existing content. While these functions offer practical benefits, they reinforce the conception of AI as a passive assistant, rather than as a cognitive or pedagogical agent capable of fostering higher-order thinking. There is a notable lack of empirical and theoretical work examining how AI might support students in developing argumentation, rhetorical awareness, disciplinary genre mastery, and reflective writing practices.

This research focuses on deeper exploration of how AI might serve as a metacognitive scaffold, namely how it can transform into a tool that supports not only *what* students write but *how they think about writing* in English as the second language. One promising avenue lies in the emerging practice of *meta-prompting*: the intentional formulation of prompts that guide AI to behave in specific ways (e.g., as a Socratic tutor, sceptical peer, or thesis advisor) to stimulate critical engagement. Yet, little has been done to examine how students learn to create such prompts, how this practice affects their writing processes, or how it might reshape the pedagogical landscape of writing instruction. Moreover, the existing literature has yet to fully consider pedagogical and ethical implications of teaching students not just to use AI, but to use it strategically—to engage in dialogic inquiry, self-questioning, and iterative refinement of ideas. The potential benefits of this approach include not only improved textual output, but also deeper metacognitive awareness, increased motivation, and the democratization of formative feedback. However, these benefits must be balanced against critical concerns about authorship, academic integrity, and over-reliance on machine-generated content.

This study addresses the following research questions:

1. In what ways can generative AI function as a *metacognitive scaffold* in writing instruction?
2. How does the integration of AI affect students' *cognitive engagement* with rhetorical and genre-based tasks?
3. What are the *pedagogical and ethical implications* of AI-supported writing processes?

By investigating these questions, the study aims to contribute to the development of a new theoretical and practical framework for AI-integrated writing pedagogy. The novelty of this investigation is in the use of AI as a dialogic scaffold that moves beyond traditional uses of AI that is focused only on grammar check or paraphrasing. Instead, students are taught to test its capacity to engage in Socratic dialogue as a form of structured, inquiry-based method of teaching where the AI (now instead of a teacher) guides students through open-ended questions rather than direct instruction, thus leading them to come to conclusions upon their own intellectual effort. The aim is not to transmit information but to stimulate critical thinking, self-reflection, and collaborative reasoning. This approach tests presumptions of Cognitive Load Theory, namely whether AI reduces or redirects cognitive load during writing tasks, and how this affects true learning versus surface-level performance.

2. LITERATURE REVIEW OF AI IN WRITING PEDAGOGY

The integration of artificial intelligence into writing instruction has rapidly advanced in the wake of generative language models like ChatGPT (OpenAI), Claude (Anthropic), and Gemini (Google DeepMind). These tools can produce coherent, structured text in

response to prompts, perform real-time paraphrasing, and simulate conversational feedback. Early pedagogical applications have focused on AI functional affordances—its ability to support vocabulary development, correct grammar, or provide summaries (El Ebyary & Windeatt, 2019). This is AI writing instruction that offers feedback and support, scoring and evaluating students' work automatically (Shermis M. D. and Burstein J., 2003). AI technological solutions like Intelligent Essay Assessor, Pigai, Grammarly, Wordtune, Writerly, and Google Docs, deliver scores immediately and provide diagnostic feedback relating to text organization, grammar, language use (Chen C. E. and Cheng W. E., 2008). This feature makes them suitable for both formative and summative assessment, and provides for a self-regulated learning environment (Ariyanto M. S., Mukminatien N., and Tresnadewi S., 2021; Zhanga L. and Huang Z., H. J. So, 2020). These features that align with students' autonomous work are visible in Writerly and Google Docs as they offer services such as Grammar Improver, Elaborator, Synonym, Sentence Booster, Convincing Bullet Points, Essay Assistant, Essay Outline (Seyyedrezaie Z. S., Ghonsooly B., Shahriari H., Fatemi A. H., 2016; Wale, B. D., Kassahun, Y. F., 2024; Chen, C., & Gong, Y., 2025). These uses address syntactic accuracy rather than rhetorical or conceptual sophistication. Consequently, empirical studies have found that students perceive AI tools as helpful for editing and brainstorming (Firat, 2023), while instructors express mixed feelings, citing both opportunities for scaffolding and threats to academic integrity (Kasneci et al., 2023; Mhlanga, 2023). These studies collectively indicate that AI tools, when used thoughtfully, can provide real-time feedback.

While a few studies (e.g., Dwivedi Y. K. et al., 2023) suggest that students can use AI to generate outlines or receive preliminary feedback on logical coherence, these uses are often incidental rather than instructionally guided. Furthermore, little is known about how AI might support metacognitive reflection—students' ability to monitor, evaluate, and adjust their writing processes. A growing number of researchers argue for repositioning AI not as a content generator but as a "thinking partner" (Holmes, W., Tuomi, I., 2022). This shift entails leveraging AI to ask clarifying questions, challenge assumptions, or simulate audience responses. Such dialogic interaction may foster deeper engagement with academic discourse, provided that students are taught how to engage with the AI dialogically and critically. Thus, Kim, T. W., & Tan, Q. (2023) explore how text-generating AI can be transformed into a writing tutor that prompts critical thinking. The authors developed a prototype AI writing-support tool that encourages users to engage more deeply with their writing. Lin Z. (2023) introduces a human-AI collaborative framework that outlines effective prompting techniques and strategies for integrating AI into the writing process. The use of AI to foster higher-order writing skills, including argument development, rhetorical awareness, and genre conventions, remains a largely untapped frontier.

Finally, the ethical landscape surrounding AI use in writing is deeply complex. Questions of authorship, originality, and transparency, continue to spark debate in both academic and public forums (Clarke, 2023, Hadan et al. 2024). From a pedagogical perspective, there is a tension between enabling students to use powerful tools and ensuring that their cognitive and ethical development is not outsourced to machines. Scholars emphasize the importance of cultivating AI literacy (Long & Magerko, 2020), including the ability to question outputs, understand model limitations, and reflect on the role of AI in knowledge creation. Without reflecting guidance into these issues students risk becoming passive recipients of AI suggestions rather than critical users capable of shaping their writing through thoughtful engagement. The existing literature suggests that

while generative AI has clear potential to enhance writing instruction, its current use is primarily focused on lower-order concerns. There remains a significant research gap in understanding how AI can be positioned as a metacognitive scaffold, particularly through the use of strategic and meta-level prompting, which this study seeks to address.

3. THEORETICAL FRAMEWORK

Cognitive Load Theory (CLT), originally proposed by John Sweller (1988), is a widely accepted theoretical framework in educational psychology that explains how the design of instructional materials and tasks can either facilitate or hinder learning. Grounded in the principles of human cognitive architecture, particularly the limitations of working memory and the vast capacity of long-term memory, CLT has significant implications for instructional design, especially in complex domains such as writing, STEM education, and problem-solving tasks. CLT distinguishes between different types of cognitive load that affect how learners process and internalize new information. Among these, intrinsic and extraneous cognitive loads are the most central. Managing these cognitive loads effectively through instructional scaffolding is critical for ensuring meaningful and lasting learning. Intrinsic cognitive load refers to the inherent difficulty associated with the material being learned. It is determined by the element interactivity of the content—the number of elements that must be simultaneously processed, as well as their degree of interconnectedness. This type of load is irreducible, in that it reflects the complexity of the subject matter relative to the learner's prior knowledge. While intrinsic load cannot be eliminated, it can be managed through techniques such as: sequencing from simple to complex tasks, segmenting information into manageable units, providing examples to reduce the cognitive burden during early learning stages. Extraneous cognitive load results from poor instructional design that distracts or misdirects the learner's limited working memory resources. It is not tied to the learning content itself but to the way information is presented, or how the tasks are structured.

Scaffolding, as integrated into instructional design by CLT scholars, refers to the temporary support provided to learners to perform tasks they could not complete independently. In CLT terms, scaffolding is a powerful means to reduce extraneous load and optimize intrinsic load by guiding learners through complex tasks. Scaffolding can take many forms, such as: step-by-step guides, templates and sentence frames, AI-driven dialogic tools that simulate tutor's questioning. Sweller et al. (1998) emphasized the importance of scaffolding in designing cognitive load-sensitive instruction, where the learner's stage of expertise determines the level and type of support. Early-stage learners benefit most from high guidance, whereas more advanced learners can handle problem-based learning without cognitive overload. Academic writing is a cognitively demanding task involving high element of interactivity—planning, organizing, synthesizing sources, attending to genre conventions, and maintaining coherence. Instructional design that fails to support these demands risks overloading the student's working memory. Scaffolding in writing instruction may include: writing outlines, AI-generated Socratic dialogue prompts, rubrics and exemplars, progressive tasks from sentence-level to discourse-level writing. By aligning scaffolds with CLT principles, instructors can ensure that students allocate cognitive resources to essential learning processes rather than unnecessary distractions. Vygotsky's sociocultural theory of learning and the concept of the Zone of Proximal Development (ZPD) have significantly

reshaped educational paradigms across disciplines. It emphasizes that cognitive development occurs not in isolation but through meaningful interaction with others, with language playing a central mediating role. Instructional support or scaffolding within ZPD is essential. It allows learners to accomplish tasks they could not manage alone and facilitates the internalization of new strategies, concepts, and modes of thinking.

Writing, from a sociocultural standpoint, is not merely a cognitive activity but a social practice embedded in specific communities, disciplines, and genres. Bakhtin (1981) has emphasized that all writing is inherently dialogic—it responds to, builds upon, or positions itself against the words of others. This dialogic view sees writing as a way of entering into conversation with scholarly discourses, adopting disciplinary norms, and negotiating meaning within epistemic communities. In this light, learning to write is not simply a matter of mastering grammar or structure, but of participating in a discourse community and learning to make meaning through socially sanctioned forms of communication. Tools like AI can serve as dialogic partners, simulating the guidance of a “more capable other” within the learner’s ZPD. Dialogic learning, informed by Vygotsky and extended by scholars like Mercer (2000) and Alexander (2008), foregrounds the importance of dialogue in constructing knowledge. This approach emphasizes reciprocity, open-ended questioning, cumulative reasoning, and shared control over learning tasks. In writing instruction, dialogic teaching methods include Socratic questioning (Stojković, N., Zerkín, D. (2023)), and reflective dialogue around genre, audience, and purpose. These strategies cultivate metacognition and critical thinking, as students are not only writing but also learning to articulate the rationale behind their rhetorical and structural choices. The sociocultural theory also provides a compelling lens through which to view the pedagogical use of AI-based writing tools. When designed dialogically, generative AI systems can act as mediators, guiding students through zones of proximal writing development. Rather than merely correcting surface-level errors, such systems can be prompted to engage in metacognitive questioning, simulate peer feedback, thus offering a form of interactive scaffolding that aligns with Vygotskian principles. This use of AI in the ZPD raises both possibilities and ethical questions. In this respect, the teacher’s role shifts from knowledge-transmitter to dialogue facilitator and AI-literacy coach, helping students frame strategic prompts, evaluate AI feedback, and maintain epistemic agency.

Finally, drawing on Paulo Freire’s critical pedagogy, the use of AI as a scaffold in teaching academic writing can be seen both as a dialogic and emancipatory practice. Freire emphasized the importance of learner agency, critical reflection, and the co-construction of knowledge, rejecting the “banking model” in which students are passive recipients (*Pedagogy of the Oppressed*, 1970). In a similar vein, AI—when used not as a static evaluator but as a responsive, interactive tool—can serve as a partner that engages students in an iterative writing process. Through personalized feedback, questioning prompts, and guided revision strategies, AI encourages learners to think critically about their arguments, structure, and rhetorical choices. This interaction fosters critical consciousness, as students begin to see writing not merely as a technical skill but as a mode of inquiry and transformation. In this framework, AI does not replace the teacher but complements a pedagogy rooted in empowerment and dialogic engagement.

In order to responsibly integrate AI as a dialogic scaffold, it is essential to adopt a framework rooted in dialogic pedagogy. Such a framework should incorporate the following components:

Table 1 Framework of theoretical basis for AI implementation

Component	Theoretical basis	AI implementation
Exploratory Dialogue	Mercer (2000)	Prompt AI to ask open-ended, clarifying, and counter-argument questions
Reciprocal Interaction	Alexander (2008)	Students revise their writing based on AI questions and re-engage with new prompts
Scaffolded ZPD Engagement	Vygotsky (1978)	AI helps students attempt tasks slightly beyond their current competence
Critical Evaluation	Freire (1970); metacognition literature	Students are taught to evaluate AI responses critically, not accept them passively

This framework shifts the focus from using AI as a convenience tool to engaging with it as a partner, one that must be approached with reflection, structure, and criticality. AI as a dialogic agent represents both an opportunity and a challenge for writing instruction. Rooted in Vygotsky's theory of mediated learning and extended through Mercer's (2000) and Alexander's (2008) dialogic frameworks, AI has the potential to enrich the writing process by simulating the kinds of peer and tutor interaction that fosters deep engagement, reflection, and growth. However, this promise can only be realized through pedagogical intentionality: students must be taught to prompt strategically, evaluate critically, and remain epistemically active. In doing so, we move toward a model of writing instruction that is not only technologically enhanced, but dialogically and ethically grounded.

4. TEACHING PROMPTING AND METAPROMPTING – CLASSROOM EXAMPLE

Central to AI pedagogical utility is the prompt—the linguistic input that shapes the output of a generative model. Yet, the pedagogical implications of teaching students how to prompt, and more importantly, how to do so strategically, remain under-theorized. Prompting is not simply a technical skill but a cognitive and rhetorical act: the way a prompt is framed can determine whether AI reinforces surface-level understanding or catalyzes critical thinking. Recent scholarship has begun to explore the concept of meta-prompting—designing prompts that direct AI to adopt specific roles (e.g., sceptic, reviewer, tutor), or to engage in particular pedagogical styles (e.g., Socratic questioning) (Zhang, Y., Yuan, Y., Yao, A. C. C. (2023); Suzgun, M., Kalai, A. T. (2024)). Research in computer-supported collaborative learning (CSCL) suggests that such prompts can foster deeper reflection and argument revision. However, this approach has not been widely examined in the context of writing instruction. Designing prompts to elicit Socratic-style dialogue from AI requires intentional phrasing that encourages the model to question back or guide thinking, rather than simply provide answers.

In order to illustrate this practice here is a skeleton of a classroom activity process in teaching writing for technical purposes using AI tools. Teacher begins by introducing the method and immediately invokes students meta cognition. Example: *“How have you used AI tools when writing technical reports or lab summaries? What worked? What didn't?”* This can be a brief whole-class discussion to activate prior experience and raise awareness of AI as a tool. Then, students are provided with a weak paragraph on an electronics topic (e.g., “The use of IoT sensors in smart grids”) and are asked to analyze what the paragraph lacks (e.g., vague claims, unclear structure, poor technical justification),

and choose their goal (e.g., improve clarity, enhance argumentation). Then, students draft a strategic AI prompt based on the goal, for example, “Act like a Socratic tutor and help me improve this paragraph by asking critical questions about clarity and technical justification.”

Here is a concrete example of how students engaged AI as a Socratic dialogic partner during writing course practice (upon teacher’s instruction on how to prompt strategically). This investigation was conducted in spring 2025 at Harbin Institute of Technology, Harbin, China, with thirty third year students whose English language proficiency is between upper intermediate to advanced proficiency level. The context was the task to write an overview of new software strongholds and to formulate strong arguments in favour.

Student: "I need to write an overview of X software features. I don't want just to enumerate them, I want to make original arguments that clearly show this software is superb. I want you to ask me questions in a Socratic way that will lead me to formulate those arguments precisely."

To this ChatGPT responded:

"Excellent approach, using Socratic questioning can really help uncover the deeper value behind the features. Let's begin by focusing on what truly makes this software stand out."

First question:

What specific problem or inefficiency does your software solve better than any existing alternative—and why does that matter deeply to the people who will use it?

*Take a moment to reflect not just on the feature, but the **transformational impact** it has on the user's experience or workflow."*

Student: Can you give me hints on how to expand these points?

ChatGPT: Here are the hints for you to expand:

- *This software eliminates repetitive manual tasks, streamlining workflows and saving users hours each week.*
- *By addressing pain points that competitors ignore, this software empowers users to achieve more in less time, improving their productivity.*
- *Imagine instantly seeing the impact of software-generated analytics without doing any extra work—it's a game changer for decision making.*

Student: I wish to dig deeper into these claims, how would you direct me?

ChatGPT: Let's dig deeper into the these issues:

- *Why is removing the "extra work" so critical in the context of analytics—and what kind of decisions become faster, more accurate, or more confident because of this?*
- *Also, consider: Who suffers most from current analytics friction? What emotional or strategic relief does this "instant visibility" provide?*

*This will help you articulate not just that it's faster, but why that speed translates to **real competitive advantage**.*

Follow up:

- *Explain how the instant visibility helps users avoid common pitfalls in decision making.*
- *Highlight the tangible advantages gained by decision-makers who are freed from lengthy data analysis.*
- *Reason how the software minimizes opportunities for errors in decision making by removing the need for manual calculations.*

This example demonstrates ChatGPT is capable of producing structured, question-driven dialogue that encourages critical reflection rather than passive content consumption, it helps students move from description to argumentation (which is an often mistake and hurdle in this type of activity as students find it difficult to differentiate between the two), and most importantly creates space for epistemic agency by guiding students to actively build their claims by inquiry. The point to remember is that AI tools do not inherently utilise Socratic model, but they need to be explicitly instructed to do so. Both teachers and students need to bear this in mind as AI is default to providing direct answers unless prompted otherwise. So if a student or teacher wants to engage the AI in a Socratic or inquiry-based dialogue, the prompt should explicitly frame that expectation. By stating it in the prompt, AI gets a role and method to follow, “Socratic” signalling that the user wants questions, not answers. Without this explicit prompt, AI as usual, offers advice, summaries, or a thesis statement, albeit short-circuiting the critical thinking process. In this *strategic AI prompting*, the prompts are *meta prompts* which have significant pedagogical implications as students enhance their reasoning skills. Also, another novelty lies in the fact that now the ability to meta-prompt (i.e., instructing the AI on how to interact) is part of academic literacy in AI-mediated environments and one of the most important pedagogical shifts emerging with AI in writing instruction. Educational implications are profound. As noted, students learn how to interact with AI productively by knowing what type of help they need, instructing the AI how to give that help, and reflecting on whether the AI output aligns with their thinking goals. As for teaching implications, this clearly demonstrates that teaching prompting and meta prompting becomes part of academic literacy curriculum, especially in writing courses. It is important to make sure students comprehend the benefit of strategic prompting, rather than just using the AI well, is all about improving their rationalising skills. This whole process is visually summarized in Fig. 1.

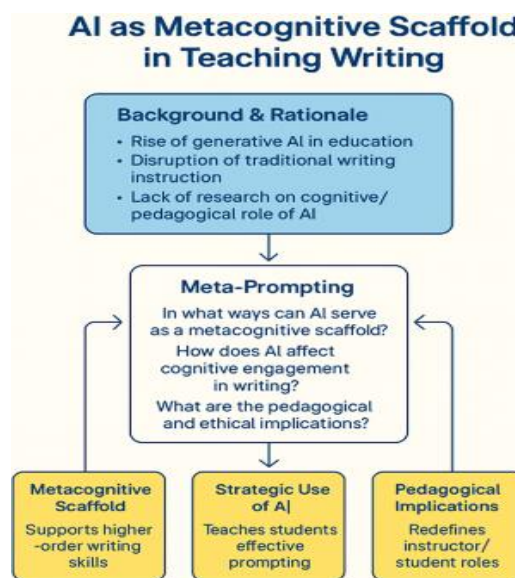


Fig. 1 Use of AI as metacognitive scaffold for writing instruction

Here are some examples of students' actual interaction with AI, their own meta prompting.

Table 2 Students' meta prompting according to writing goals

Writing goal	Prompt type	Example in context
Clarify structure	Socratic questioning	"Ask me questions to help clarify the logical flow of my paragraph about induced behaviour in children."
Improve argument	Critical peer	"Challenge the strength of my claim that nuclear energy production is more efficient than wind sourcing."
Explain concept better	Explainer	"Help me explain the principle of sound modulation in simpler language for a non-specialist audience."
Compare studies	Analytical reviewer	"Help me compare the methodologies of these two studies on energy efficiency."

After this activity they revise their writing. The critical phase in the activity is reflection. It can be expanded into sharing the experience among the class. For this research purpose, students were given a reflection worksheet with these questions: What kind of prompt did you use? How did the AI respond? What changes did you make to your paragraph? How did this help your understanding of the topic or writing strategy? In order to enhance the impact of the practice one may ask students to submit for evaluation both their original and revised versions, the prompts used, and their short commentary on experience. Based on students' answers teacher gains insight into their reflection quality, and whether they fully grasp the task by checking prompt clarity and purpose alignment. Meta prompts matter as they activate higher-order thinking (analysis, synthesis, evaluation), prevent passive use of AI (e.g., copy-paste answers), support metacognitive awareness—students reflect on *how* they learn or write, not just *what* they produce, and enable dialogic learning—AI becomes a *thinking partner*, not a solution provider.

In the teaching practice, teachers and students can develop prompt typology to guide their strategic use. Teachers can develop this further to make meta prompting templates for different stages of writing, and for different written formats.

Table 3 Prompt templates by writing goals

Your goal	Prompt type	Example prompt
Clarify technical structure	Socratic questioning	Ask me questions to help clarify the logic of my paragraph on the working principle of local governments
Improve argument strength	Critical peer review	Challenge my claim that nuclear energy production is better than wind energy sourcing
Simplify complex explanations	Explainer for non-specialists	Help me explain psychological testing in simple language for someone new to this field
Compare research findings	Analytical reviewer	Help me compare two papers on energy efficiency. Ask me what's similar and different.
Identify missing evidence	Evidence checker	What kind of evidence would make my argument about smart grid security stronger?
Improve flow or cohesion	Structure coach	Read my paragraph on healthy food and ask me where transitions or logic are unclear.

In class one can use live prompting demonstration to show how a particular type of meta prompt produces certain type of feedback. Then, invite students to share their prompts and AI outputs, analyse them together in terms of strong and weak points of the prompts as reflected in the feedback received. Students can also have prompts revision practice to compare effective and ineffective prompts. This should precede students' reflecting on how AI influenced their thinking, their intentions and outcomes after each AI interaction (e.g., via short journals or think-alouds).

An important benefit of teaching students to prompt strategically, to ask better questions, is that they are less likely to default to letting AI do the work for them. They stop perceiving it as an answer machine, but as a thinking tool, since inquiry and not efficiency is emphasized. This is the process that achieves results through practice. Moving from prompting to meta prompting can be summarized in the following way:

Table 4 Levels of prompting: from basic to metacognitive

Level	Prompt type	Goal	Example
1. Basic	Direct Instruction	Get fast answers or summaries	"Define Ohm's Law."
2. Functional	Task Completion	Complete a writing or editing task	"Edit this paragraph to improve grammar."
3. Strategic	Cognitive Support	Support specific writing challenges	"Help me clarify the logic in this paragraph about adolescents' behavior."
4. Dialogic	Interactive Thinking	Use questions to challenge or guide reasoning	"Ask me questions about the assumptions in my argument on energy efficiency."
5. Metacognitive	Self-Aware Prompting	Reflect on process, assumptions, goals	"Act like a critical thinking coach. Help me identify weaknesses in my argument structure and ask me what I could do to improve clarity."

In order to fully benefit from this strategic questioning of AI it is paramount that after the practice the teacher invites students to self-reflect by using the same method – appealing to their meta cognition.

5. RESEARCH

In order to validate this approach Design-Based Research (DBR) methodology of research was applied, integrating repeating cycles of design, implementation, analysis, and refinement within authentic educational contexts (Reeves, 2006). Design-based research is methodology that employs theories of learning that research how learning ecologies, intricate, multidirectional web of interactions among professors and students, as well as students among themselves, impact students' learning (Peffer and Renken, 2016). DBR was selected to explore how generative AI can function as a dialogic scaffold in writing instruction, allowing for responsive adaptation to real-world pedagogical challenges. The research was conducted at Harbin Institute of Technology (HIT), Harbin, China, in spring 2025, where the course on English is taught to thirty

third-year Bachelor students (same for both institutions), over a six-week period. It was approved by the Ethics committee of HIT, all thirty students submitted their written consent, and none dropped out.

5.1. Data collection

To capture both the process and outcomes of AI-mediated writing instruction, the study employed a multi-modal data collection strategy that integrated both qualitative and quantitative instruments.

Pre- and post-intervention writing samples. At the beginning and end of the study, all thirty students produced writing tasks in their disciplinary context (argumentative essays or short technical reports). This resulted in a total of 60 samples (30 pre-intervention and 30 post-intervention). Samples were evaluated using a standardized writing rubric (see Appendix A), covering argument clarity, coherence, structure, use of sources, genre awareness, and linguistic accuracy. Two independent raters—the course instructor and an external ESP specialist—scored all samples. Inter-rater reliability was established using Cohen’s kappa, which yielded a reliability coefficient of $\kappa = .82$, indicating substantial agreement. Discrepancies were discussed and resolved through consensus.

Think-aloud protocols. During live writing sessions, students participated in concurrent think-aloud protocols based on Ericsson and Simon’s (1993) framework. Students verbalized their thoughts, strategies, and decisions while interacting with AI (ChatGPT). A pilot trial with three students was conducted to validate the procedure, ensuring consistency and feasibility. Think-aloud sessions were audio-recorded, transcribed, and thematically coded for metacognitive strategies, affective responses, and instances of frustration or reliance. The excerpt of training script and prompts are provided in Appendix B.

Semi-structured interviews and focus groups. After the intervention, all students participated in individual semi-structured interviews, and also in five focus groups of six students each so that all thirty participants were included. Interviews explored students’ perceptions of AI as a dialogic partner, trust in AI-generated feedback, ethical considerations, and self-efficacy as writers. Focus groups examined social dimensions of AI use, including peer prompt-sharing, integrity concerns, and negotiation of AI’s role in the writing process. The interview guides and focus group agendas are presented in Appendix C.

AI interaction logs. All prompts and AI responses were automatically captured and anonymized. These were coded for prompt type (directive, exploratory, evaluative), dialogic depth, and evidence of iterative refinement. Interaction logs complemented the qualitative data by providing a record of students’ evolving prompting strategies. AI interaction logs captured every student prompt and the corresponding AI response during the six-week intervention. These logs were anonymized and systematically coded using NVivo. Coding categories included prompt type (directive, exploratory, evaluative), dialogic depth (surface-level vs. reflective questioning), and iteration patterns (e.g., single-turn vs. recursive refinement). Frequency counts and progression mapping were applied to track how students shifted from basic to metacognitive prompting over time. Triangulation with think-aloud protocols allowed validation of whether logged strategies reflected genuine cognitive engagement resulting in students’ evolving prompting strategies - the gradual progression in how students framed their inputs to AI across the intervention. At the start, many relied on basic prompts (“fix grammar in this paragraph”). By mid-intervention, students increasingly used strategic prompts (“ask me

questions about the weaknesses of my counterargument”), and toward the end, they demonstrated metacognitive prompts (“act as a critical peer reviewer; help me identify assumptions in my reasoning and ask me to defend them”). For example, one student’s trajectory moved from:

- Week 1: “Summarize this text more clearly.”
- Week 3: “Help me clarify the logical flow of my introduction.”
- Week 6: “Challenge my argumentation on X by questioning unstated assumptions.”

This developmental arc illustrates that students not only refined prompt phrasing but also deepened their rhetorical and reflective use of AI as a dialogic partner.

5.2. AI intervention

The intervention was designed as a modular AI-supported writing program, embedded within the existing English language curriculum. AI was integrated at key stages of the writing process. Students received instructor-guided AI usage protocol (see Appendix D) provided students with structured guidelines for responsible and reflective AI engagement, outlined recommended stages of use, emphasized ethical literacy, strategic prompting, and critical evaluation of AI outputs. The instructional framing positioned AI as a cognitive collaborator, not a shortcut, and included explicit guidance on distinguishing between permissible support and academic misconduct. The intervention was designed to introduce generative AI as a dialogic scaffold in academic writing, strategically integrated across multiple stages of the writing process. The goal was not to automate writing tasks, but to foster metacognitive engagement, enhance rhetorical awareness, and support genre-specific literacy. This pedagogical design was implemented over a six-week period and iteratively refined according to Design-Based Research principles.

AI was embedded into three distinct writing stages to maximize cognitive benefit and minimize over-reliance.

a. Planning and brainstorming

Students used generative AI (ChatGPT) to generate topic ideas, pose preliminary research questions, and outline argument structures. Prompts included models of the following types:

- “Suggest three potential arguments for a position paper on sustainable energy in electronics.”
- “Help me frame a research question relevant to multimedia design in education.”

This phase targeted idea generation and topic clarification, and supported students in moving from vague to specific academic objectives.

b. Outlining and structural mapping

Students prompted AI to assist in mapping out essay or report structures. AI was asked to simulate the genre conventions of academic texts (e.g., IMRaD for reports, or Toulmin structure for essays). Example prompt:

- “Based on this thesis statement, outline a 4-paragraph structure with topic sentences and logical transitions.”

This phase emphasized macro-organization, cohesion, and logical flow of arguments.

c. Revision and dialogic feedback

AI was repurposed as a peer-like dialogic agent, offering Socratic prompts, counterarguments, and suggestions for elaboration or clarity. Students were trained to engage critically with AI feedback and to iterate based on both automated and human feedback.

- *Student: “Is my counterargument strong enough?”*
- *AI: “Your counterargument mentions an alternative view, but could you elaborate on why this view might be compelling to some readers? What evidence supports that view?”*

This phase fostered reflective revision and critical thinking.

5.2.1. Instructor-Guided AI Usage Protocol

To ensure responsible and pedagogically sound integration of AI, a structured usage protocol was developed. It included instruction on prompt engineering, including examples of exploratory, strategic, and meta-prompts; a framework for ethical AI use that emphasized academic integrity, avoidance of plagiarism, transparency about the role of AI in the writing process. There was also an emphasis on critical literacy, encouraging students to evaluate, question, and challenge AI suggestions, rather than accept them passively. Students completed reflective journals documenting how they used AI, which prompts were effective, and how AI feedback influenced their revisions. These reflections were later analyzed for metacognitive depth.

The instructor functioned as a mediator, not only guiding the technical use of AI tools, but also framing AI as a cognitive collaborator, facilitating discussions about AI limitations and bias, and providing human feedback alongside AI-generated suggestions to balance perspectives. In the particular context of this research, instructors also tailored tasks and prompts to align with disciplinary genres, ensuring that AI-assisted tasks remained grounded in content-relevant communication (e.g., technical descriptions, data commentary, cause-effect structures).

The intervention design followed a graduate scaffolding model. In first two weeks there was an obvious need for intense instructor’s support, guiding through prompting templates, and focusing on familiarization. In weeks three and four, there was observed increased student autonomy in generating and refining prompts, and in their guided reflection on feedback. In the final two weeks, there was independent use of AI for revision, and independent ability to select when and how to use AI based on task and genre. This gradual release of responsibility encouraged students to internalize prompting strategies and treat AI not as a shortcut, but as a tool for dialogic inquiry and self growth.

6. DATA ANALYSIS

The analysis proceeded on multiple levels, corresponding to the research questions. Qualitative analysis referred to thematic analysis of think-aloud data, interviews, and focus groups using NVivo qualitative data analysis software to identify emergent patterns in student attitudes, metacognitive development, and ethical awareness. Discourse analysis of AI-student interaction transcripts was also employed to evaluate the dialogic quality of prompts and responses, aligned with frameworks from Mercer (2000) and

Vygotsky (1978). Discourse analysis of AI-student interaction transcripts focused on four criteria:

1. Dialogic depth – whether AI responses prompted elaboration, justification, or reflection.
2. Reciprocity – evidence of iterative turn-taking where students reformulated their writing in response to AI's questions.
3. Critical stance – presence of evaluative moves (e.g., questioning assumptions, weighing evidence) rather than surface-level corrections.
4. Agency markers – indications that students retained authorship, such as rejecting AI suggestions, rephrasing them, or integrating them with original ideas.

These criteria allowed the analysis to determine whether interactions reflected genuine dialogic scaffolding or superficial tool use.

Quantitative analysis consisted of rubric-based scoring of pre- and post-intervention writing to measure improvements in higher-order writing skills (argumentation, clarity, genre control) and cognitive load assessment, using self-report Likert-scale surveys based on Sweller's Cognitive Load Theory (1988), to evaluate perceived task difficulty and mental effort across AI-supported and traditional writing tasks. This mixed-methods framework allowed for a comprehensive understanding of how AI functions not only as a text aid but as a metacognitive and dialogic scaffold in writing education.

Here is the procedure regarding qualitative and discourse analysis. First, students' verbal reflections during think-aloud sessions and interviews were transcribed and coded.

- *Theme*: "AI as Thinking Partner"

Quote: "It's like the AI kept asking me why I chose a certain example—I never thought about that before."

Code: Cognitive engagement, Reflection

- *Theme*: "Over-reliance"

Quote: "I just accepted the suggestion because it sounded better, even though I didn't fully understand it."

Code: Dependency, Surface-level use

These themes were mapped to research questions about metacognition and ethical engagement. AI logs were analyzed using a dialogic lens (Mercer, 2000). Prompts and responses were coded for dialogic depth.

- Student prompt: "How can I improve this introduction?"

AI response: "What is the main argument you want your readers to understand?"

Should it be made more explicit in the first two sentences?"

Analysis: AI response is a Socratic prompt → Encourages student reflection → Dialogic move.

- *Coding categories*:

- Exploratory dialogue (Open-ended questions)
- Authorial development (Metacognitive prompts)
- Procedural requests (Fix grammar – coded as low-cognitive prompt)

This allowed for analysis of how students use AI and whether interactions fostered deeper engagement.

Regarding quantitative analysis, here is the essay writing example. Essays were evaluated using an academic writing rubric (1–5 scale across categories), where statistical tests (paired t-tests) assessed significance of the observed gains.

Table 5 Quantitative analysis of writing rubric scores (pre/post intervention)

Category	Pre-Mean	Post-Mean	Δ
Argument clarity	2.5	3.8	+1.3
Structure	3.0	4.0	+1.0
Use of Sources	2.2	3.5	+1.3
Genre Awareness	2.8	3.9	+1.1

As for cognitive load self-assessment, in the semi-structured interview, students rated the perceived difficulty on a 7-point Likert scale after tasks, adapted from Sweller et al. (1998):

Example Question: “How mentally demanding was this writing task?”

- 1 = Very low load
- 7 = Extremely high load

Table 6 Cognitive load data: evidence of AI scaffolding function

Task Stage	Traditional Mean	AI-Supported Mean
Brainstorming	5.2	3.1
Revision	6.0	4.0

The quantitative data comparing cognitive load during brainstorming and revision tasks reveal meaningful insights into how AI support affects students’ mental effort. Results are interpreted to evaluate whether AI reduced extraneous load while supporting germane load. These values, derived from self-reported cognitive load measures (on a 1–7 scale), suggest that the integration of AI significantly reduced perceived cognitive effort in both stages of the writing process. In the context of Cognitive Load Theory, the drop in load from 5.2 to 3.1 during brainstorming, and from 6.0 to 4.0 during revision, indicates a reduction in extraneous cognitive load. Extraneous load refers to the effort expended on tasks that do not directly contribute to learning, often caused by unclear instructions, overwhelming complexity, or disorganized workflows. Students reported during semi-structured interviews that AI helped them reduce this type of load by offering structured prompts and questions that helped students initiate ideas during brainstorming, as well as providing clear suggestions and alternative phrasings during revision, which minimized confusion or indecision.

At the same time, students’ qualitative data (reflections sessions) showed that they used the cognitive AI support to engage more deeply in germane processing—the cognitive effort devoted to learning and logical structuring of the text. For example, during brainstorming, students focused more on refining thesis statements and arguments, rather than just finding something to write, while during revision, they spent time reworking transitions and evaluating coherence, rather than struggling with surface-level issues like sentence polishing.

All this demonstrates that AI did not eliminate cognitive effort, but *rechanneled it toward higher-order writing tasks*, which is a hallmark of effective scaffolding. These results support the claim that when thoughtfully integrated, generative AI can act as a metacognitive tool that reduces unproductive cognitive effort, while enhancing productive struggle. It helps students focus not on how to write, but on what and why they are writing—shifting the locus of effort from mechanical to rhetorical concerns. This underscores AI potential to serve as a

pedagogically meaningful support, not merely a time-saver or convenience. Realizing this potential depends on how AI is introduced, guided, and reflected upon within the classroom context.

7. DISCUSSION

The findings of this study align with the transformative potential of generative AI when conceptualized not as an automation tool but as a cognitive and dialogic scaffold within writing instruction. By embedding AI strategically across the writing process, particularly at the planning, outlining, and revision stages, students engaged in deeper metacognitive reflection, showed improved rhetorical awareness, and demonstrated gains in genre-specific competencies. Consistent with the theoretical framing rooted in Cognitive Load Theory, the intervention appeared to reduce extraneous cognitive load, while increasing germane load, allowing students to focus on higher-order concerns, such as argument development and audience awareness. Students' use of exploratory and metacognitive prompts ("How can I make this argument more nuanced?") illustrates that with guided training, they were able to activate AI as a thinking partner, not merely a corrective device. This aligns with previous studies that have advocated AI as a support mechanism for idea generation and feedback (Zhai, 2022), but extends the field by providing empirical evidence that students can be taught to prompt AI in ways that stimulate critical thinking. Students who engaged most effectively with AI were those who developed strategic prompting literacy—a skill that should now be foregrounded in writing pedagogy.

Building on sociocultural and dialogic learning theories, this study reveals that AI, when properly prompted, can simulate dialogic interaction typically associated with peer feedback or tutoring. The data analysis of AI interaction logs showed that students frequently revised their texts in response to AI-generated questions or challenges, demonstrating a level of reciprocal engagement. However, while AI succeeded in simulating certain types of feedback, it lacked true epistemological grounding and sometimes returned generic or overly cautious suggestions. These limitations reaffirm the need for human mediation, especially to teach students how to evaluate AI output critically, a finding in line with calls for critical AI literacy in educational settings (Dehouche, 2021).

While the use of AI stimulated meaningful learning in many cases, ethical concerns emerged. Some students admitted to using AI for entire sections of their writing, revealing a gray area between scaffolding and substitution. This reinforces the argument that AI integration must be ethically framed and transparently discussed, particularly in assessments where originality and attribution are essential. Instructors must help students navigate the affordances and limits of AI—an instructional need not yet well represented in current academic integrity policies. Moreover, the role of the instructor proved to be essential. Students benefitted most from AI when its use was embedded in a guided, purposefully designed curriculum, rather than left to ad hoc experimentation. Instructor-led instructions on prompt engineering, critical evaluation, and genre awareness were pivotal in determining whether AI acted as a dialogic agent or simply a surface-level editor.

8. CONCLUSION - REFRAMING WRITING INSTRUCTION

The integration of generative AI into writing pedagogy necessitates a profound shift in instructional design. Rather than viewing AI as a shortcut or threat to authentic learning, educators are encouraged to reframe it as a cognitive tool, one that supports higher-order thinking, rhetorical awareness, and metacognitive engagement. This reframing aligns with sociocultural theories of learning, in which knowledge construction occurs through interaction and mediation. AI, when strategically prompted, can function not as an answer-generator but as a dialogic scaffold, facilitating reflection, revision, and genre-specific reasoning. This presents both opportunities and imperatives for the future of pedagogy in higher education. Traditional writing instruction has often relied on linear processes—prewriting, drafting, revising—with instructor feedback delivered at fixed intervals. With AI capable of on-demand, interactive, and recursive dialogue, writing instruction can evolve toward a more fluid and responsive model, where students co-construct meaning in real time, guided by AI as a dialogic partner. This shift may redefine the teacher's role—not as a sole source of expertise, but as a *metacognitive coach and ethical mediator*. Writing assessment must be reimagined to account for AI-augmented authorship. Rubrics and evaluation strategies will need to distinguish between authentic intellectual effort (e.g., strategic prompting, revision decisions) and passive AI dependency (e.g., outsourcing entire sections without reflection). This calls for process-based assessment models that emphasize drafting, reflection, and engagement with feedback, including AI feedback, over purely final-product evaluation. This pedagogical shift requires instructor mediation to ensure students develop the AI literacy and ethical discernment necessary to navigate this new learning environment. Teachers must help students understand when and how to engage AI, how to critically assess its outputs, and how to preserve their epistemic agency. This is particularly critical in preventing dependency, superficial engagement, or violations of academic integrity.

This research contributes to theory in two critical ways. First, it extends the concept of scaffolding, traditionally understood as human support within a learner's Zone of Proximal Development to include non-human, dialogic agents. While Sweller et al. (1998) focused on instructional design to reduce extraneous cognitive load, the use of AI introduces a dynamic, adaptive scaffold that responds to learner input in real time, blurring the boundaries between static resources and interactive mediation. Second, the study offers insights into how AI reconfigures the writing process itself. Instead of viewing writing as a solitary or linear act, AI enables a recursive, exploratory, and feedback-rich environment. This dialogic orientation redefines the student's relationship to text: writing becomes a conversation with the self and the machine, guided by prompts, counter-prompts, and iterative refinement.

Hereby, hedging needs to be employed as the implementation of AI in writing instruction is highly context-dependent. Factors such as academic discipline, student background, instructor attitudes, and institutional policy shape the ways AI is adopted and its effects on learning. Moreover, the rapid evolution of AI tools during the study period introduces a layer of volatility and raises important questions about the longevity and replicability of findings. What is possible in one semester may become obsolete or dramatically enhanced in the next. Generative AI holds transformative potential in writing instruction when it is approached as a strategic, metacognitive scaffold. To realize this potential, pedagogical design must prioritize critical awareness, ethical framing, and dialogic engagement, ensuring that students do not simply consume AI outputs but use them to interrogate, explore, and refine their thinking.

Future research should pursue longitudinal studies to examine how sustained exposure to AI-mediated writing affects skill development over time. Additionally, there is a need to explore disciplinary differences in AI adoption, as writing conventions and expectations vary widely across fields. Finally, ensuring equity of access, both in terms of digital infrastructure and AI literacy, should be central to the responsible and inclusive integration of AI into writing education.

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APPENDICES

Appendix A. Writing Rubric

Each dimension rated on a 1–5 scale (1 = very weak, 5 = excellent).

Criterion	1 (Very Weak)	2 (Weak)	3 (Adequate)	4 (Good)
Argument Clarity	Argument absent or incoherent	Vague or inconsistent	Argument stated but underdeveloped	Clear, logical argument with minor gaps
Structure & Coherence	No organization, lacks flow	Minimal structure, unclear progression	Some logical flow with weak transitions	Well-organized, mostly coherent
Use of Sources & Evidence	No evidence or misused	Minimal/irrelevant evidence	Some relevant sources, inconsistently integrated	Consistent use of relevant sources
Genre Awareness	Ignores genre conventions	Limited awareness	Basic genre features evident	Mostly appropriate genre use
Linguistic Accuracy	Frequent errors impede meaning	Many errors, distracting	Occasional errors, mostly clear	Few errors, minor lapses

Appendix B. Think-Aloud Protocol Materials

Training Script (adapted from Ericsson & Simon, 1993):

- “As you write and interact with AI, please say everything that comes into your mind.”
- “Do not explain to me, just verbalize your thoughts as they occur.”
- “If you are silent for more than 10 seconds, I will remind you to keep talking.”

Sample Prompts for Students:

- “What are you trying to achieve with this sentence?”
- “Why did you accept or reject the AI’s suggestion?”
- “What difficulties are you noticing as you write?”

Validation:

A pilot session with 3 students was conducted. Recordings were reviewed to confirm that the protocol elicited consistent verbal data without disrupting the writing task. Minor adjustments were made before full implementation.

Appendix C. Semi-Structured Interview and Focus Group Guides

Interview Agenda (individual, 25–30 min):

1. Can you describe your experience using AI during the writing process?
2. In what ways did AI help you improve your writing?
3. Were there times when you felt over-reliant on AI?
4. How much did you trust AI-generated feedback?
5. Did your confidence as a writer change during the study? If so, how?
6. What ethical concerns, if any, did you have when using AI?

Focus Group Agenda (5 groups, 6 students each, 45–60 min):

1. How did you and your peers use AI in different ways?
2. Did you share prompts or AI outputs with each other?
3. How do you think AI should (or should not) be used in academic writing?
4. What risks do you see in using AI tools for coursework?
5. What strategies helped you stay in control of the writing process, rather than outsourcing it to AI?

Appendix D: Instructor-Guided AI Usage Protocol

1. **Ethical Use Principles**
 - AI as *scaffold*, not ghostwriter.
 - Explicit acknowledgment of AI use in drafts when applicable.
 - Avoidance of verbatim copy-pasting without critical engagement.
2. **Stages of Application**
 - *Brainstorming*: Generate ideas, clarify research questions.
 - *Structuring*: Request genre-specific outlines.
 - *Revision*: Use AI for Socratic questioning and rhetorical improvement.
3. **Prompting Guidelines**
 - Start with specific goals (e.g., clarify logic, strengthen evidence).
 - Use meta-prompts to define AI's role (e.g., "*Act as a skeptical peer reviewer*").
 - Always reflect: accept, adapt, or reject AI suggestions consciously.
4. **Reflection Requirement**
 - Students submit brief journal entries on AI use: what prompt they tried, what feedback they accepted/rejected, and why.